

Full Stack Development with MERN

1. INTRODUCTION

Project Title

Book Store using MERN Stack

Team Members:

Name	Role
Shaik Jaherun	Team Member

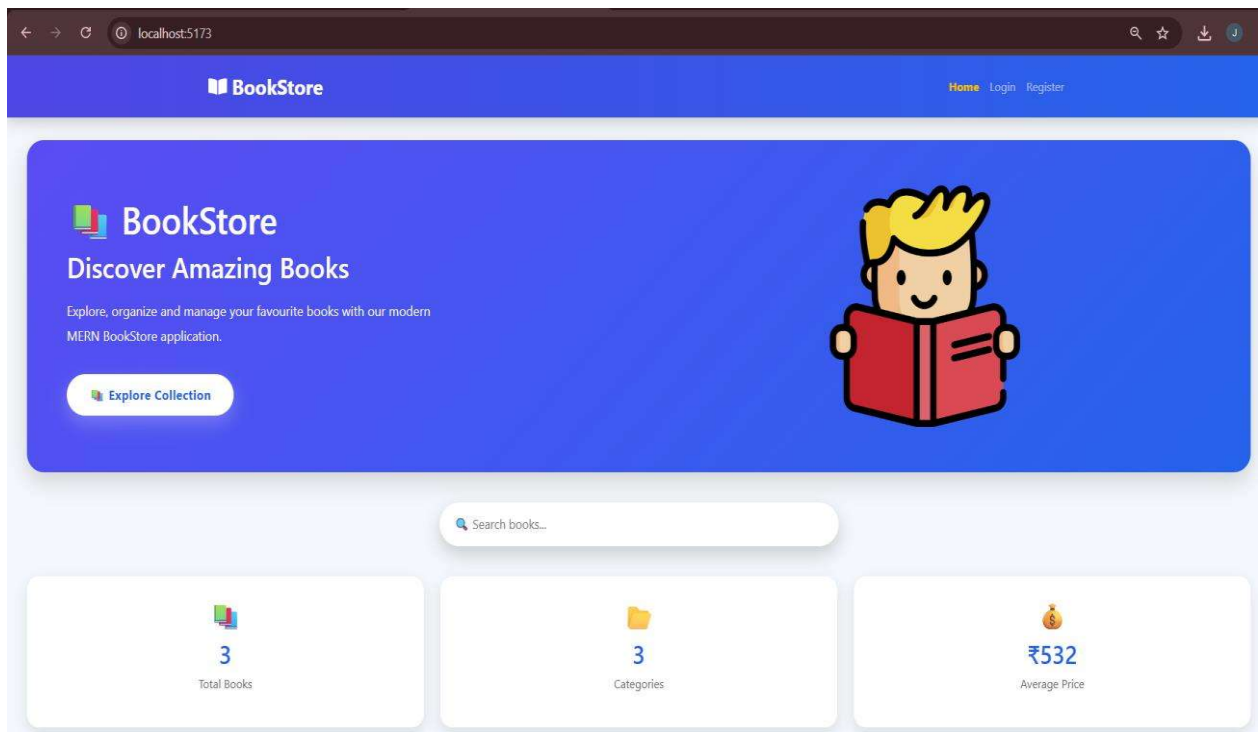
2. Project Overview

Purpose:

The **Book Store Management System** is a full-stack web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The purpose of this project is to provide an online platform where users can browse books, search by category, add books to a shopping cart, place orders, and view order history. The system also provides an administrator panel for managing books through secure CRUD operations. The project demonstrates modern web development concepts such as authentication, REST APIs, responsive UI, and database management.

Features

- User Registration & Login
- JWT Authentication
- Role-Based Authorization (Admin/User)
- Book Management (Add, Edit, Delete)
- View Book Details
- Search Books
- Category Filter
- Shopping Cart
- Checkout System
- Order Management
- Responsive Design
- SweetAlert Notifications
- Toast Notifications
- Custom 404 Error Page



3. ARCHITECTURE

Frontend

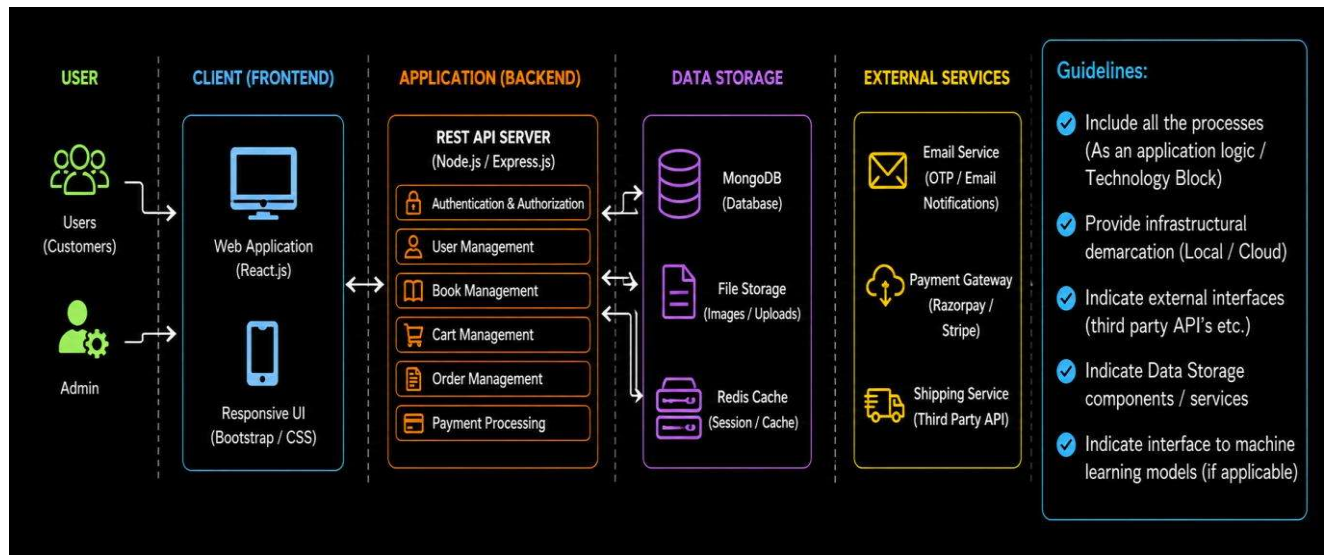
The frontend of the application is developed using **React.js**. React Router is used for page navigation, Axios is used for API communication, Bootstrap and CSS provide responsive styling, and React Toastify with SweetAlert2 improves user interaction.

Backend

The backend is developed using **Node.js** and **Express.js**. RESTful APIs handle user authentication, book management, shopping cart, and order processing. JWT authentication secures protected routes while bcrypt encrypting user passwords.

Database

MongoDB Atlas is used as the cloud database for storing user details, books, shopping cart information, and orders. Mongoose provides schema modeling and simplifies database interactions.



4. SETUP INSTRUCTIONS

Prerequisites

- Node.js
- npm
- MongoDB Atlas
- VS Code
- Git

Installation

```
git clone https://github.com/shaikjaherun/BookStore.git
```

```
cd BookStore
```

```
npm install
```

Frontend

```
cd frontend
```

```
npm install
```

```
npm run dev
```

Backend

```
cd backend
```

```
npm install
```

```
npm run dev
```

Environment Variables

PORT=5000

MONGO_URI=

mongodb+srv://bookstore:Bookstore786@cluster0.nko7ckr.mongodb.net/bookstore?retryWrites=true&w=majority&appName=Cluster0

JWT_SECRET= bookstoresecret

5. Folder Structure

BookStore/

|

|— frontend/

| |— src/

| | |— components/

| | |— pages/

| | |— services/

| | |— styles/

| | └─ App.jsx

|

|— backend/

| |— config/

| |— controllers/

| |— middleware/

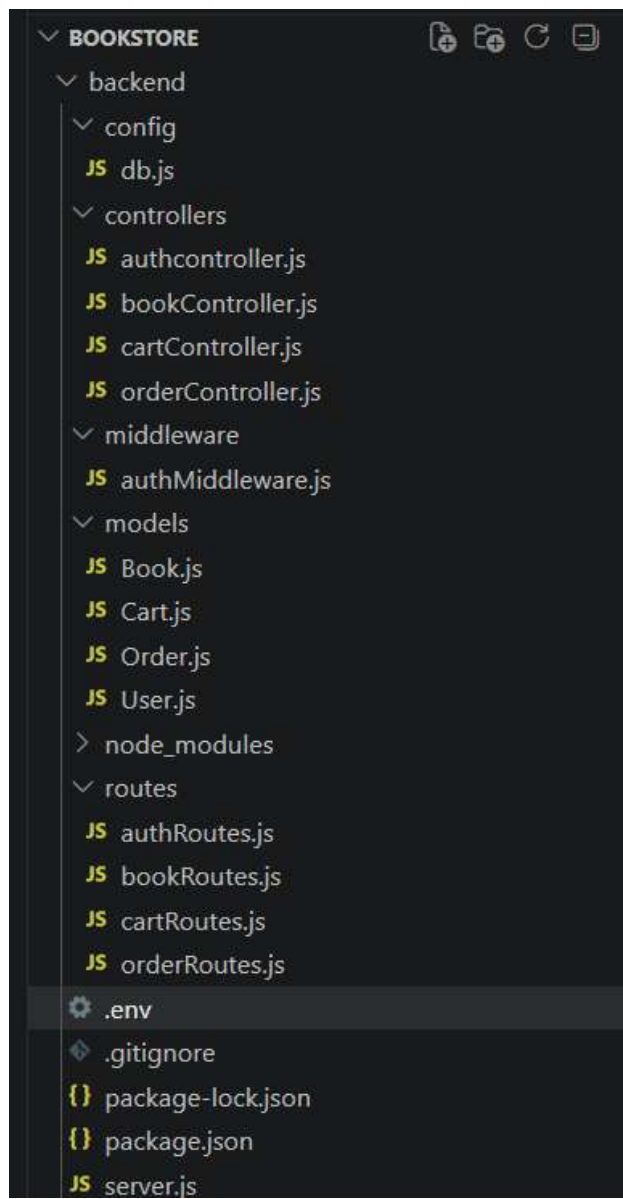
| |— models/

| |— routes/

| └─ server.js

|

└─ README.md



▼ **BOOKSTORE**

▼ frontend

➤ node_modules

▼ public

➤ images

 favicon.svg

 icons.svg

▼ src

▼ assets

 hero.png

 login.png

 react.svg

 vite.svg

▼ components

 BookCard.jsx

 Footer.jsx

 Navbar.jsx

 ProtectedRoute.jsx

▼ pages

 AddBook.jsx

 BookDetails.jsx

 Cart.jsx

 Checkout.jsx

 EditBook.jsx

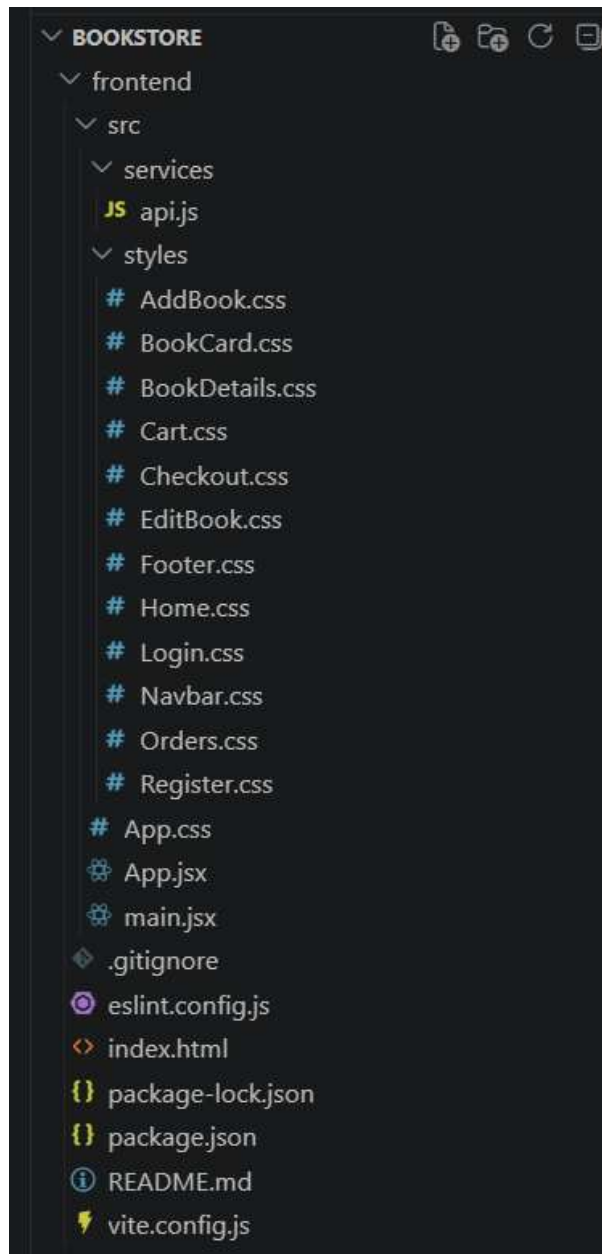
 Home.jsx

 Login.jsx

 NotFound.jsx

 Orders.jsx

 Register.jsx

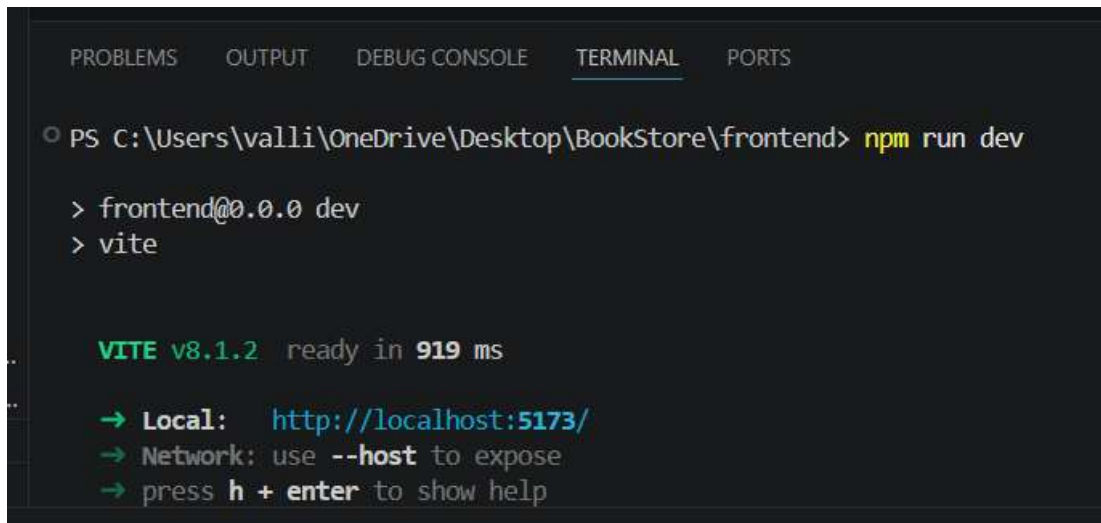


6. Running Application

Frontend

cd frontend

npm run dev



A screenshot of a VS Code terminal window with the 'TERMINAL' tab selected. The terminal shows the command 'npm run dev' being executed in the directory 'C:\Users\valli\OneDrive\Desktop\BookStore\frontend'. The output indicates that Vite v8.1.2 is ready in 919 ms and the development server is running at http://localhost:5173/. It also provides instructions on how to expose the network and how to show help.

```
PS C:\Users\valli\OneDrive\Desktop\BookStore\frontend> npm run dev

> frontend@0.0.0 dev
> vite

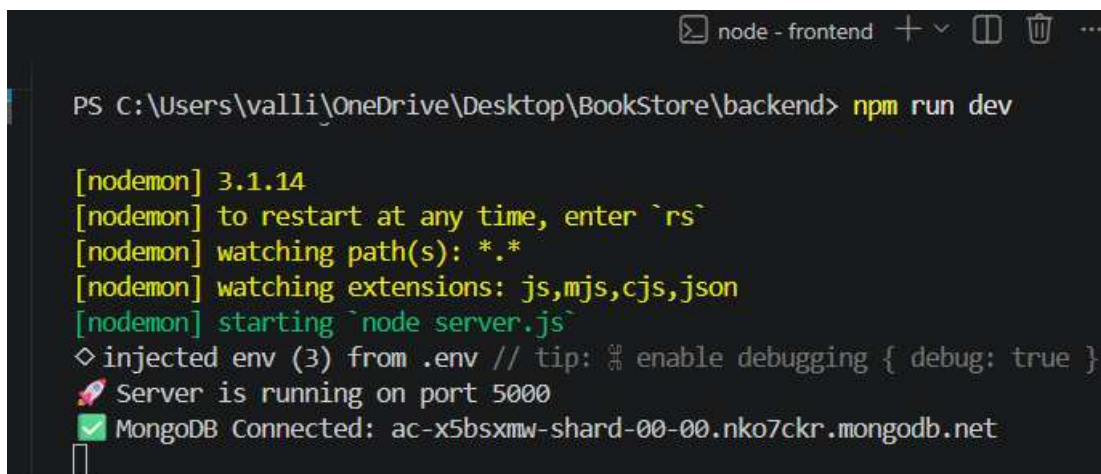
VITE v8.1.2 ready in 919 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
```

Backend

cd backend

npm run dev



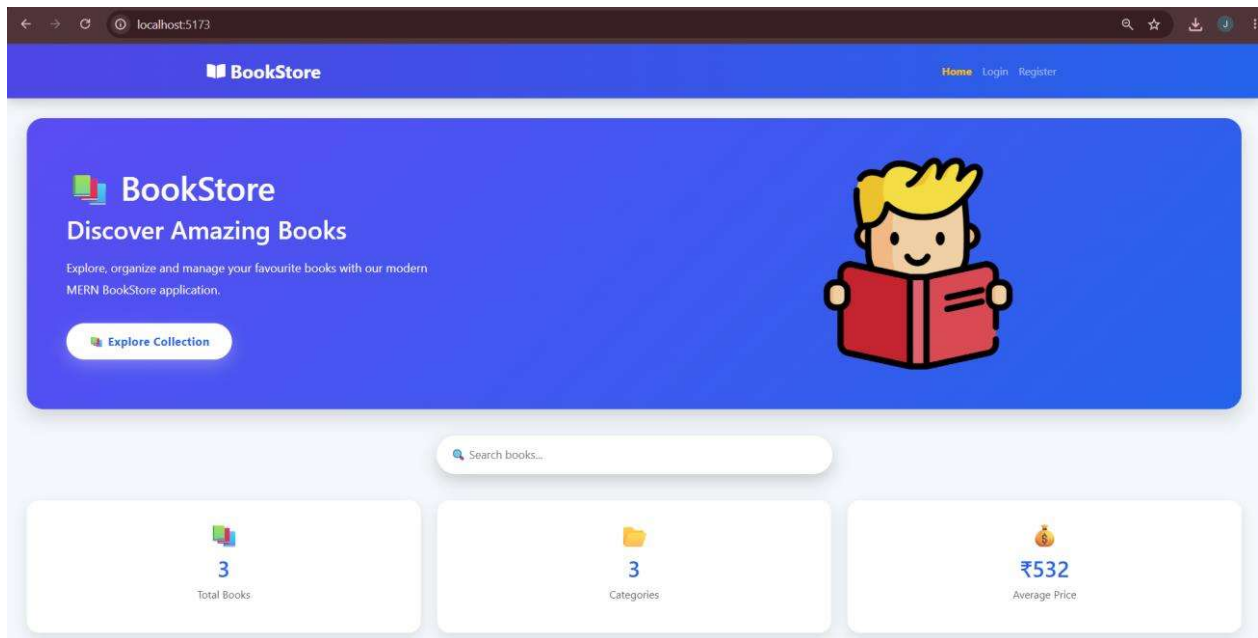
A screenshot of a VS Code terminal window with the title 'node - frontend'. The terminal shows the command 'npm run dev' being executed in the directory 'C:\Users\valli\OneDrive\Desktop\BookStore\backend'. The output shows that nodemon 3.1.14 is running, watching for file changes and starting the node server. It also shows that the environment is injected from .env, the server is running on port 5000, and MongoDB is connected.

```
PS C:\Users\valli\OneDrive\Desktop\BookStore\backend> npm run dev

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
◇ injected env (3) from .env // tip: % enable debugging { debug: true }
🚀 Server is running on port 5000
✅ MongoDB Connected: ac-x5bsxmw-shard-00-00.nko7ckr.mongodb.net
```

Open Browser

<http://localhost:5173>



7. API DOCUMENTATION

Base URL

<http://localhost:5000/api>

Authentication APIs:

1.Register User

Field	Value
Endpoint	/auth/register
Method	POST
Description	Registers a new user

Request Body

```
{  
  "name": "Shaik Jamal",  
  "email": "jamal@gmail.com",  
  "password": "123456"  
}
```

Success Response

```
{  
  "success": true,  
  "message": "User Registered Successfully"  
}
```

2. Login User

Field	Value
Endpoint	/auth/login
Method	POST
Description	Authenticates user and returns JWT Token

Request Body

```
{  
  "email": "jamal@gmail.com",  
  "password": "123456"  
}
```

Success Response

```
{  
  "success": true,  
  "token": "JWT_TOKEN"  
}
```

Book APIs

3. Get All Books

Field	Value
Endpoint	/books
Method	GET
Description	Returns all available books

4. Get Single Book

Field	Value
Endpoint	/books/:id
Method	GET
Description	Returns details of a specific book

5. Add Book (Admin)

Field	Value
Endpoint	/books
Method	POST
Description	Adds a new book

Request Body

```
{
  "title": "The Alchemist",
  "author": "Paulo Coelho",
  "price": 499,
  "category": "Novel",
  "image": "image-url",
  "description": "Book Description"
}
```

6. Update Book

Field	Value
Endpoint	/books/:id

Method	PUT
Description	Updates an existing book

7. Delete Book

Field	Value
Endpoint	/books/:id
Method	DELETE
Description	Deletes a book

Cart APIs

8. Add to Cart

Field	Value
Endpoint	/cart/add
Method	POST
Description	Adds selected book into shopping cart

9. Get Cart Items

Field	Value
Endpoint	/cart
Method	GET
Description	Returns all cart items

10. Update Cart Quantity

Field	Value
Endpoint	/cart/update/:id
Method	PUT
Description	Updates book quantity inside cart

11. Remove Cart Item

Field	Value
Endpoint	/cart/remove/:id
Method	DELETE
Description	Removes a book from cart

12. Clear Cart

Field	Value
Endpoint	/cart/clear
Method	DELETE
Description	Clears all cart items

Order APIs

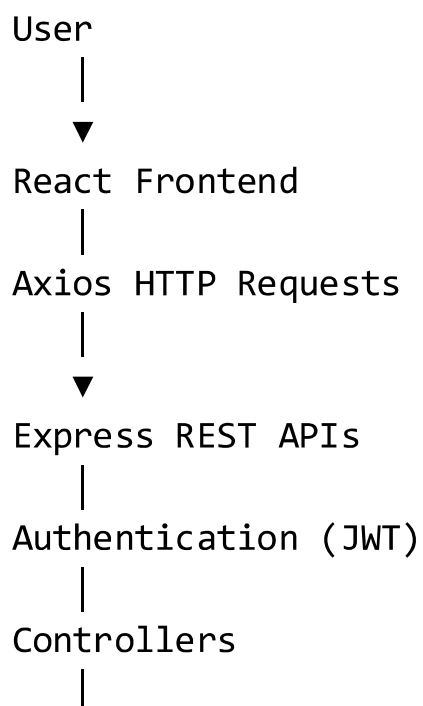
13. Place Order

Field	Value
Endpoint	/orders
Method	POST
Description	Places a new order

14. Get My Orders

Field	Value
Endpoint	/orders
Method	GET
Description	Retrieves all orders of logged-in user

API Flow



MongoDB Database



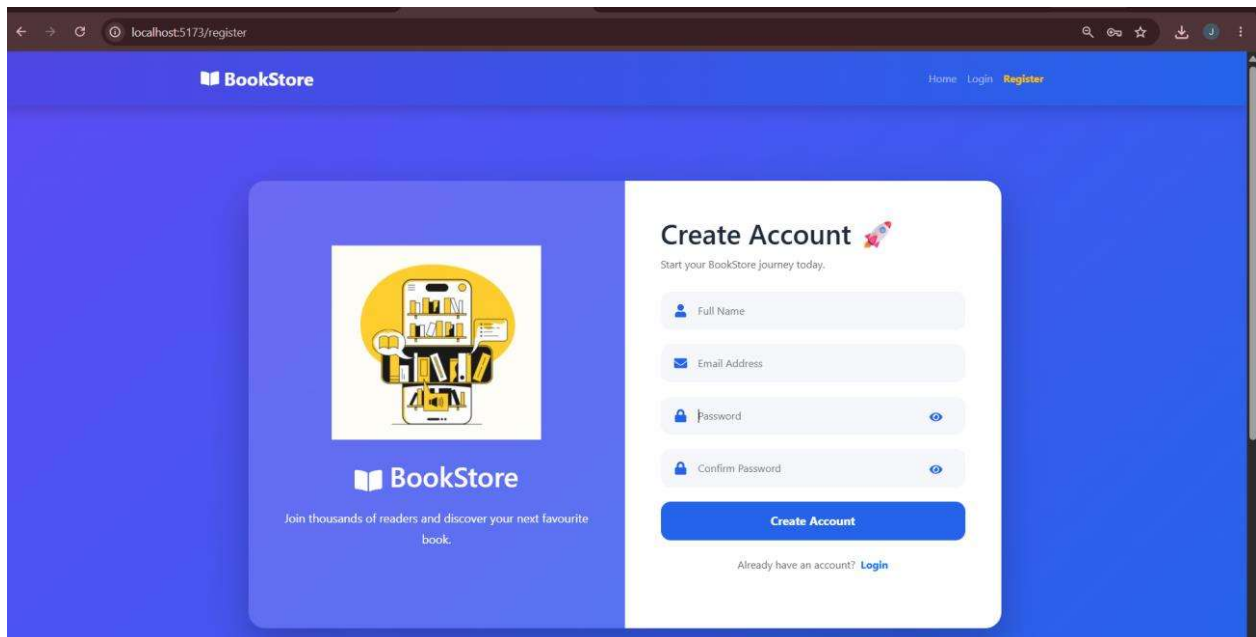
Response

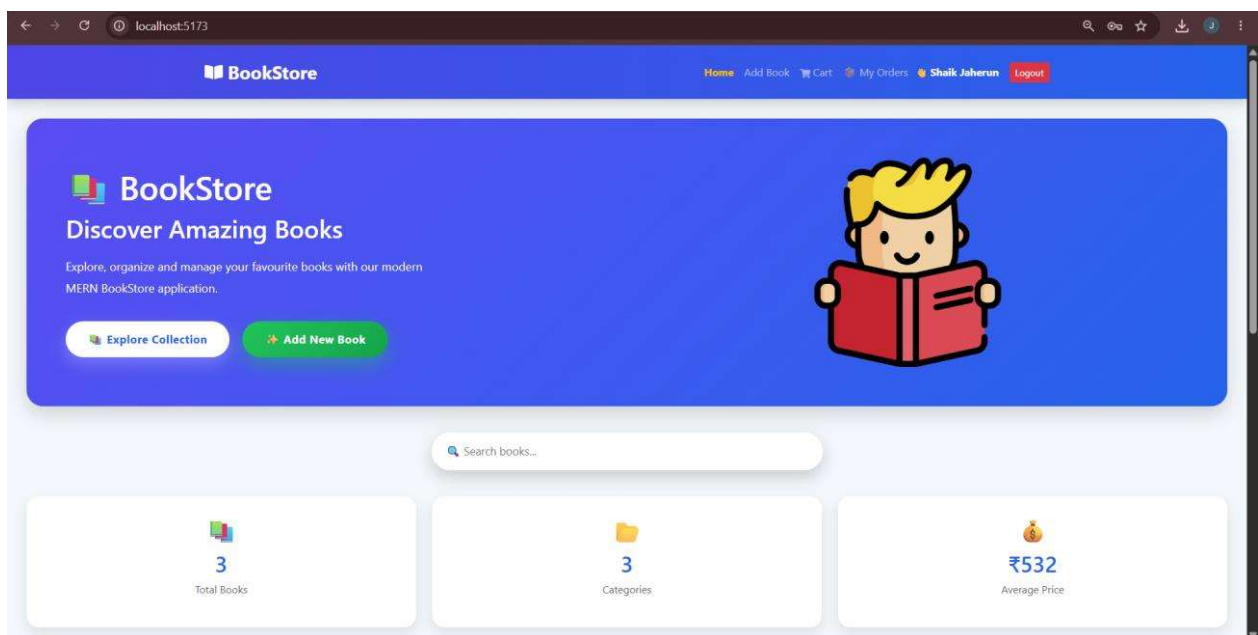
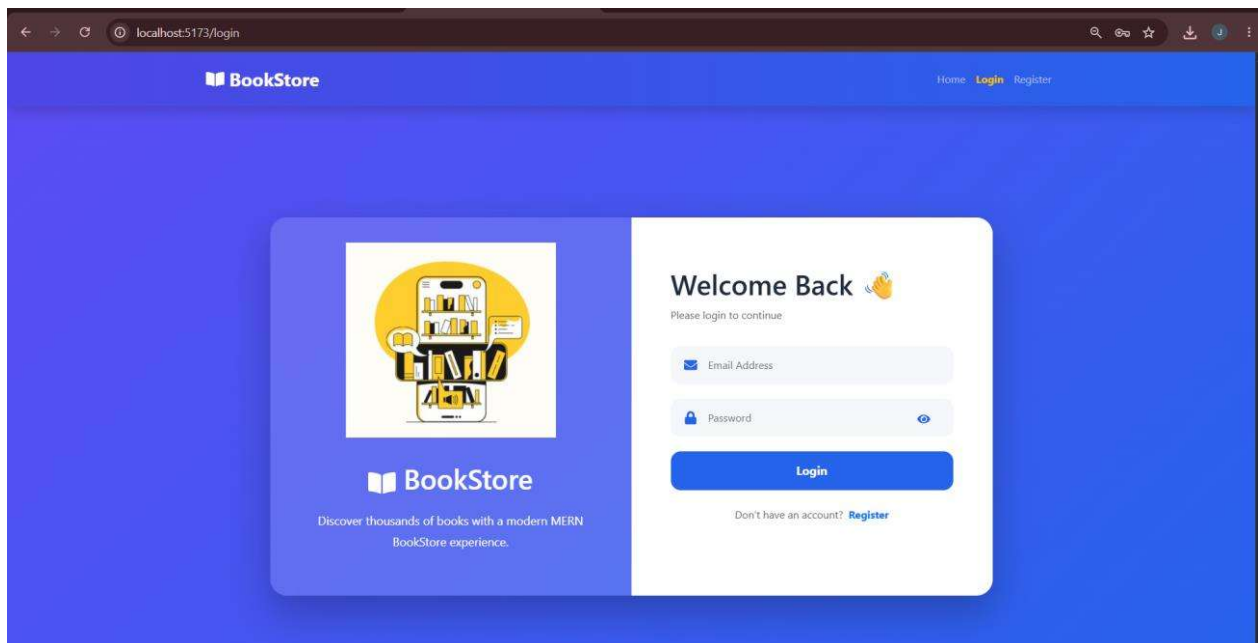


Frontend UI

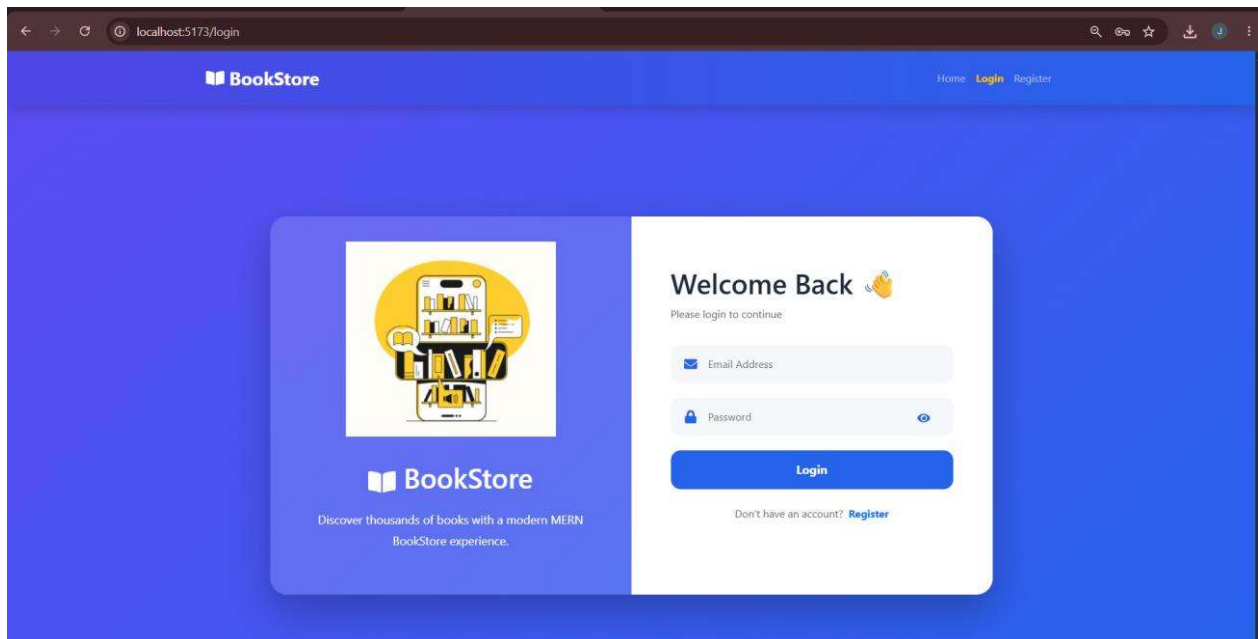
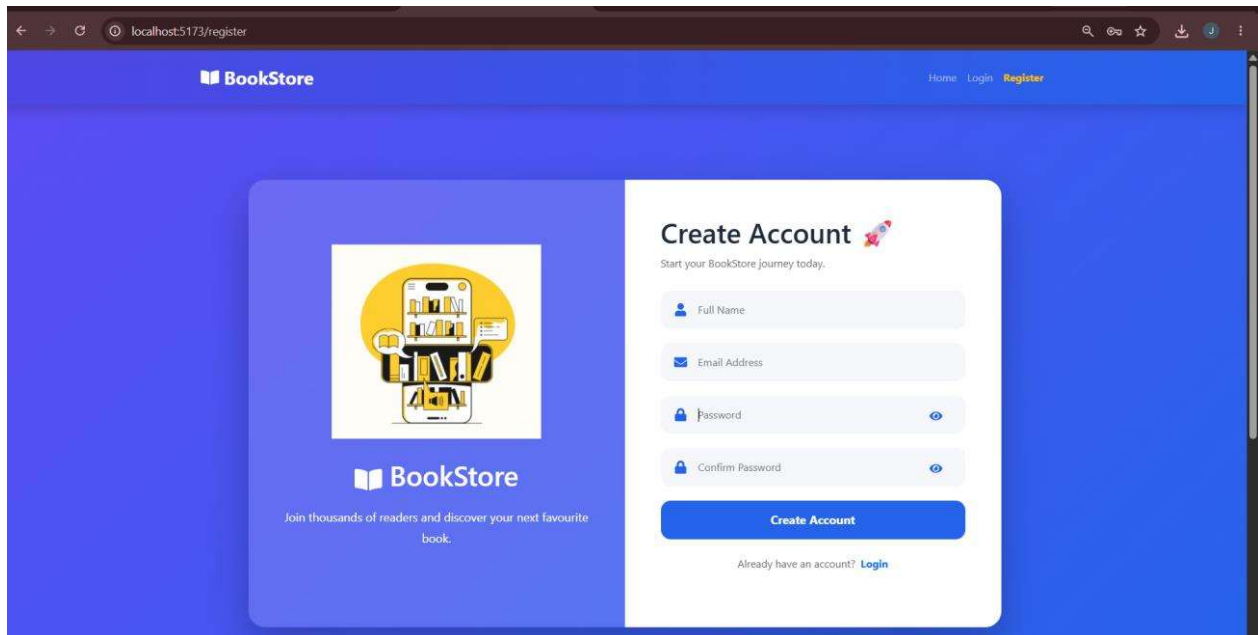
8. AUTHENTICATION

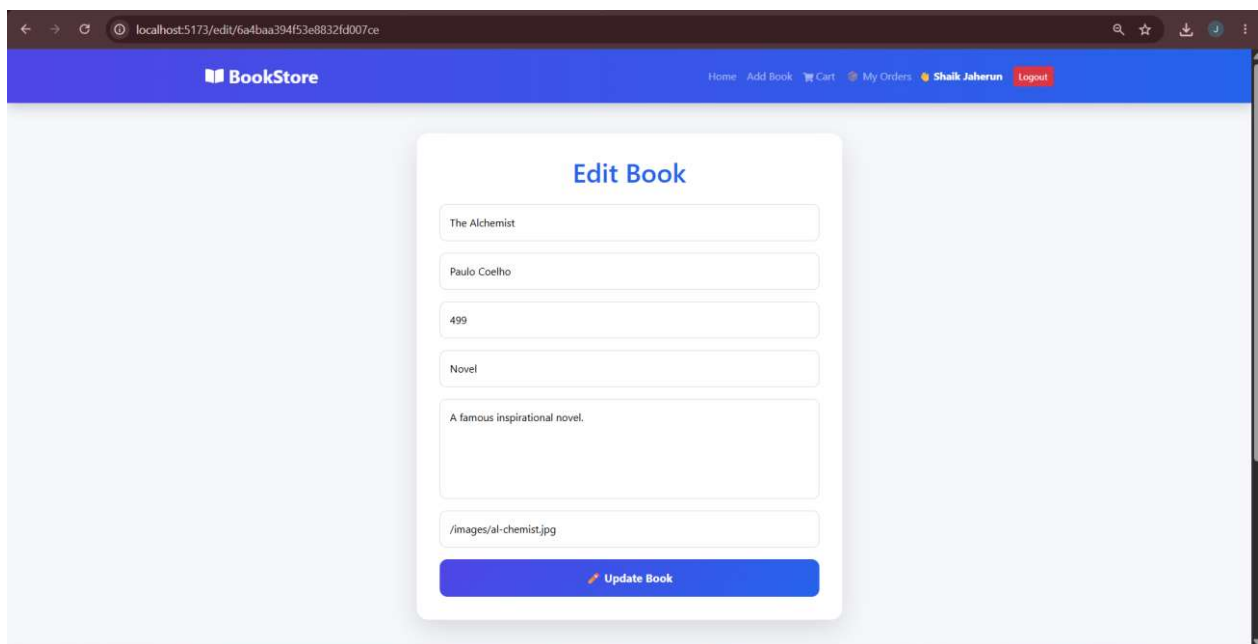
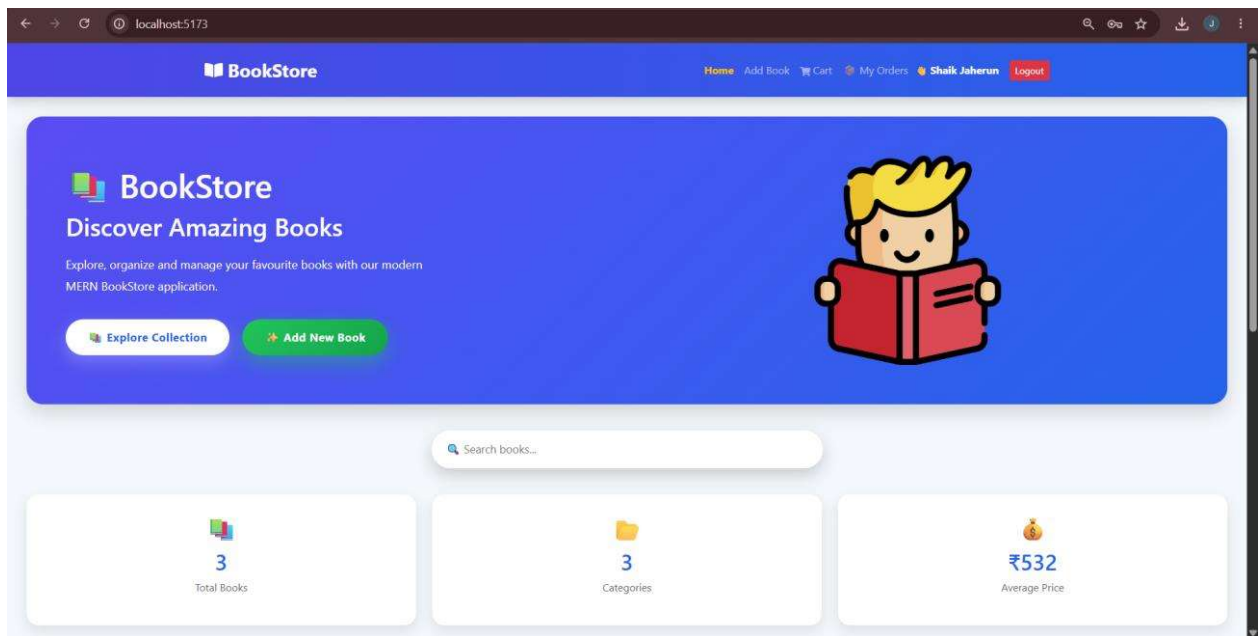
Authentication is implemented using JWT (JSON Web Token). Users register and log in using email and password. Passwords are securely hashed using bcrypt before storage. Protected routes verify the JWT token before allowing access. Role-based authorization restricts administrative functions such as adding, editing, and deleting books.





9. User Interface





← → ↻ 📄 localhost:5173/add 🔍 ☆ 📄 📄

📖 BookStore Home Add Book 🛒 Cart 📄 My Orders 🌟 Shaik Jaherun Logout

Add New Book

Book Title

Author Name

Price

Category

Book Description

Image URL

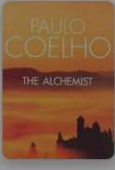
Add Book

← → ↻ 📄 localhost:5173 🔍 ☆ 📄 📄

📖 BookStore Home Add Book 🛒 Cart 📄 My Orders 🌟 Shaik Jaherun Logout

📖 Featured Books

All Novel Self-Help Personal Development



The Alchemist

Author: Paulo Coelho

Novel In Stock

★★★★★ (4.9)

₹499

A famous inspirational novel.

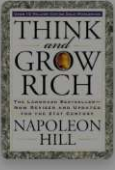
Edit Delete

!

Delete Book?

This action cannot be undone.

Yes, Delete Cancel



Think and Grow Rich

Author: Napoleon Hill

Personal Development In Stock

★★★★★ (4.9)

₹499

Success mindset

Edit Delete

📖 BookStore

Discover, manage and organize your favourite books with a modern bookstore application.

Quick Links

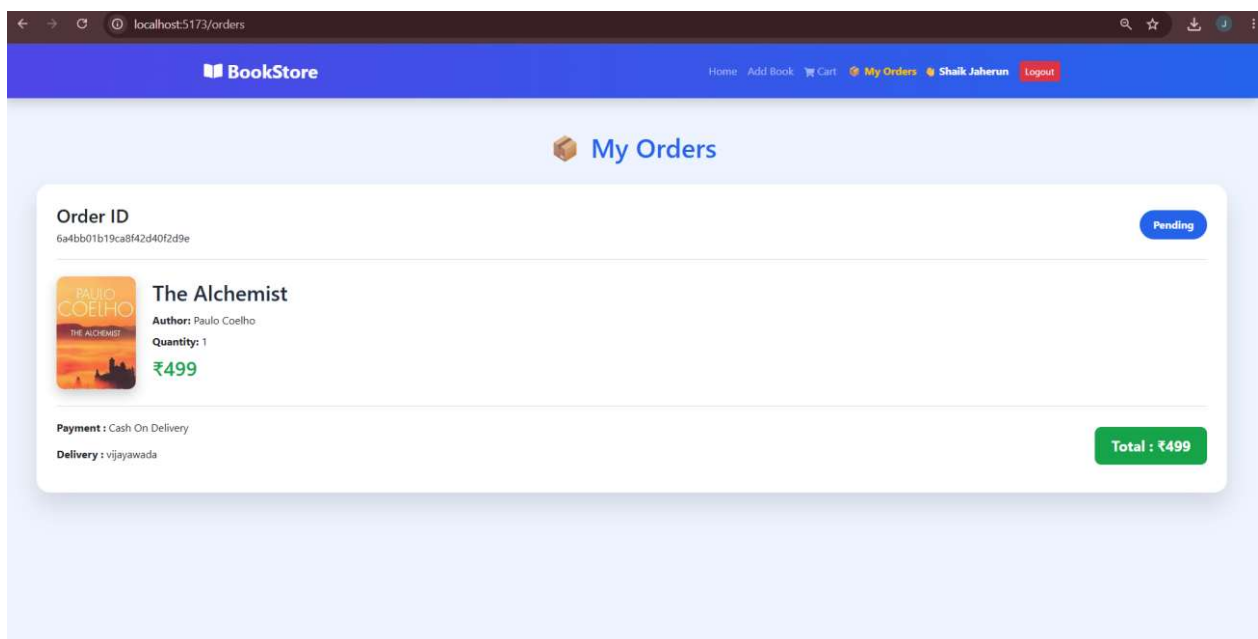
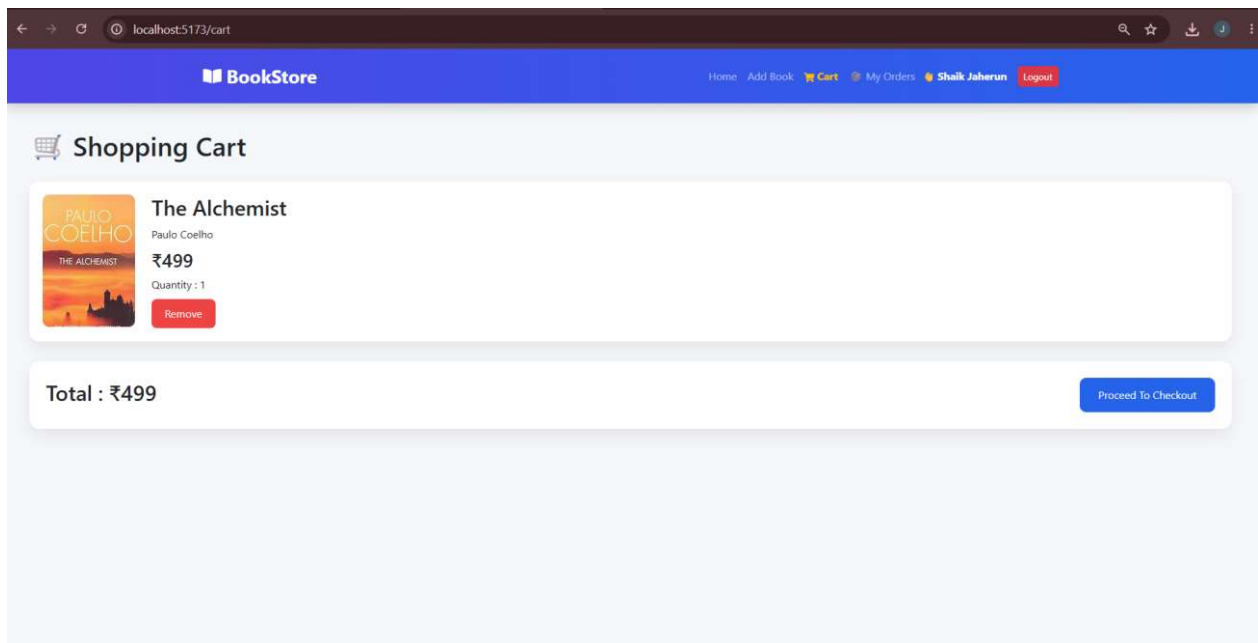
Home

Add Book

Contact

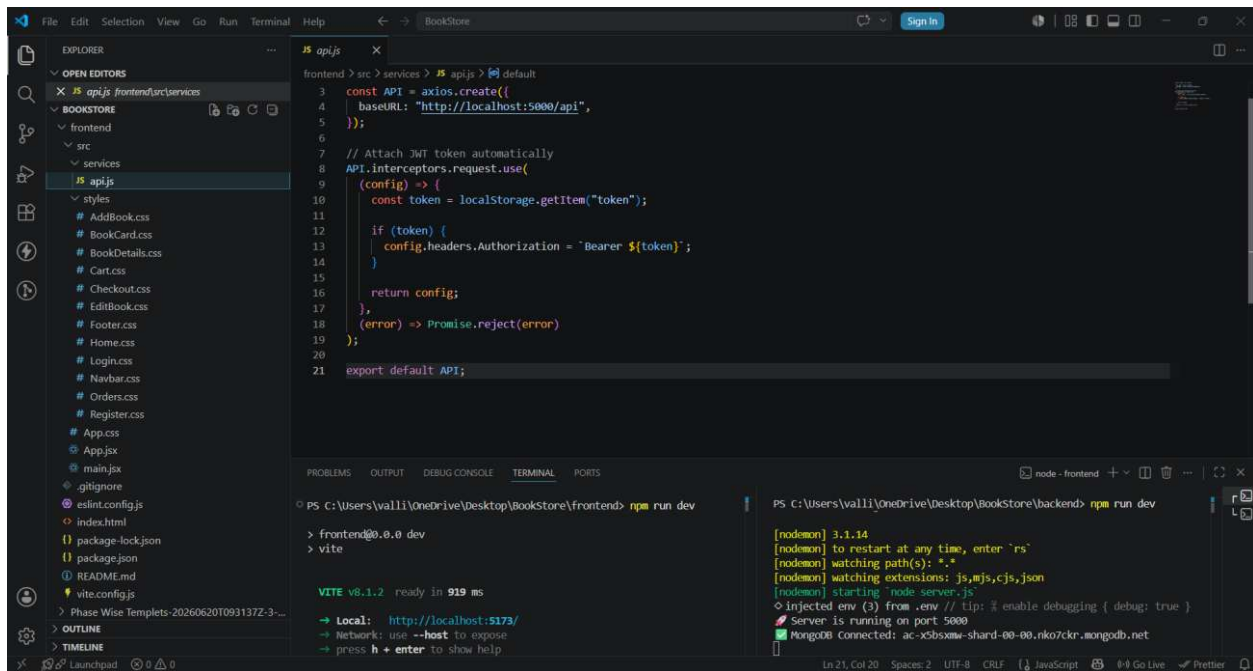
📧 bookstore@gmail.com

📞 +91 98765 43210



10. Testing

The application was tested using manual testing to verify authentication, CRUD operations, cart functionality, order processing, responsive layouts, and API integration. All major functionalities were validated successfully.



```
File Edit Selection View Go Run Terminal Help
BookStore
Sign In

EXPLORER
OPEN EDITORS
X JS api.js frontend\src\services
BOOKSTORE
  frontend
  src
  services
  JS api.js
  styles
    AddBook.css
    BookCard.css
    BookDetails.css
    Cart.css
    Checkout.css
    EditBook.css
    Footer.css
    Home.css
    Login.css
    Navbar.css
    Orders.css
    Register.css
  App.css
  App.jsx
  main.jsx
  .gitignore
  eslint.config.js
  index.html
  package-lock.json
  package.json
  README.md
  vite.config.js
  Phase Wise Templates-20260620T093137Z-3-...
OUTLINE
TIMELINE

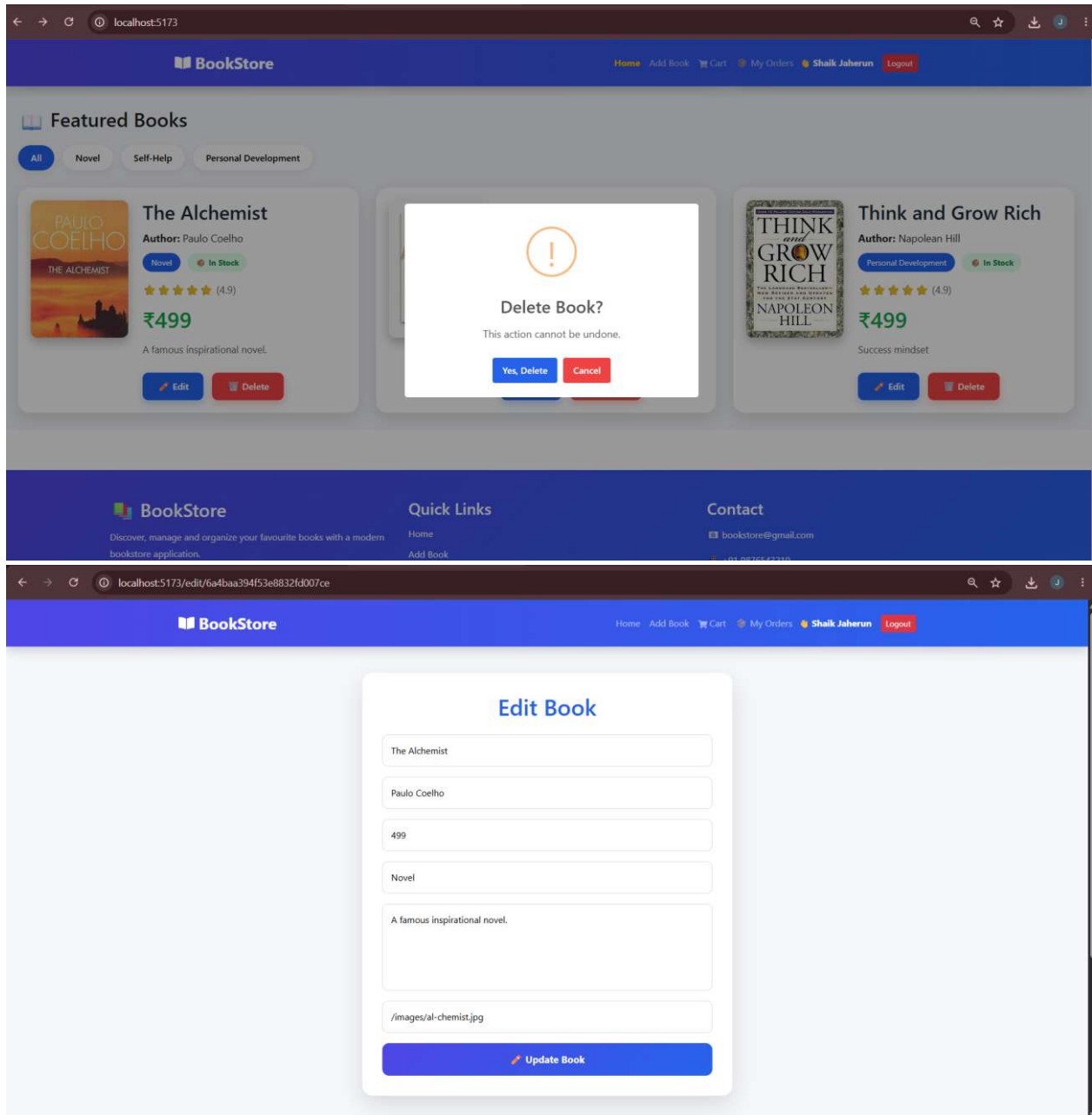
api.js
3 const API = axios.create({
4   baseURL: "http://localhost:5000/api",
5 });
6
7 // Attach JWT token automatically
8 API.interceptors.request.use(
9   (config) => {
10     const token = localStorage.getItem("token");
11
12     if (token) {
13       config.headers.Authorization = `Bearer ${token}`;
14     }
15
16     return config;
17   },
18   (error) => Promise.reject(error)
19 );
20
21 export default API;
```

```
PS C:\Users\valli\OneDrive\Desktop\BookStore\frontend> npm run dev
> frontend@0.0.0 dev
> vite

VITE v8.1.2 ready in 919 ms
  ➔ Local:   http://localhost:5173/
  ➔ Network: use --host to expose
  ➔ press h + enter to show help

PS C:\Users\valli\OneDrive\Desktop\BookStore\backend> npm run dev
[nodemon] 3.1.14
[nodemon] to restart at any time, enter 'rs'
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node server.js'
◊ injected env (3) from .env // tip: enable debugging { debug: true }
Server is running on port 5000
MongoDB Connected: ac-x5bsxmw-shard-00-00.nko7ckr.mongodb.net
```

11. Screenshots and Demo:



← → ↺ 📍 localhost:5173/orders

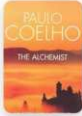
📖 BookStore

Home Add Book 🛒 Cart 📄 My Orders 🌟 Shaik Jaherun Logout

📦 My Orders

Order ID
6a4bb01b19ca8f42d40f2d9e

Pending



The Alchemist

Author: Paulo Coelho

Quantity: 1

₹499

Payment : Cash On Delivery

Delivery : vijayawada

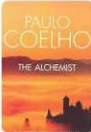
Total : ₹499

← → ↺ 📍 localhost:5173/cart

📖 BookStore

Home Add Book 🛒 Cart 📄 My Orders 🌟 Shaik Jaherun Logout

🛒 Shopping Cart



The Alchemist

Paulo Coelho

₹499

Quantity : 1

Remove

Total : ₹499

Proceed To Checkout

BookStore

[Home](#) [Add Book](#) [Cart](#) [My Orders](#) [Shaik Jaherun](#) [Logout](#)

Add New Book

[Add Book](#)

BookStore

[Home](#) [Add Book](#) [Cart](#) [My Orders](#) [Shaik Jaherun](#) [Logout](#)



BookStore

Discover Amazing Books

Explore, organize and manage your favourite books with our modern MERN BookStore application.

[Explore Collection](#) [Add New Book](#)





3

Total Books



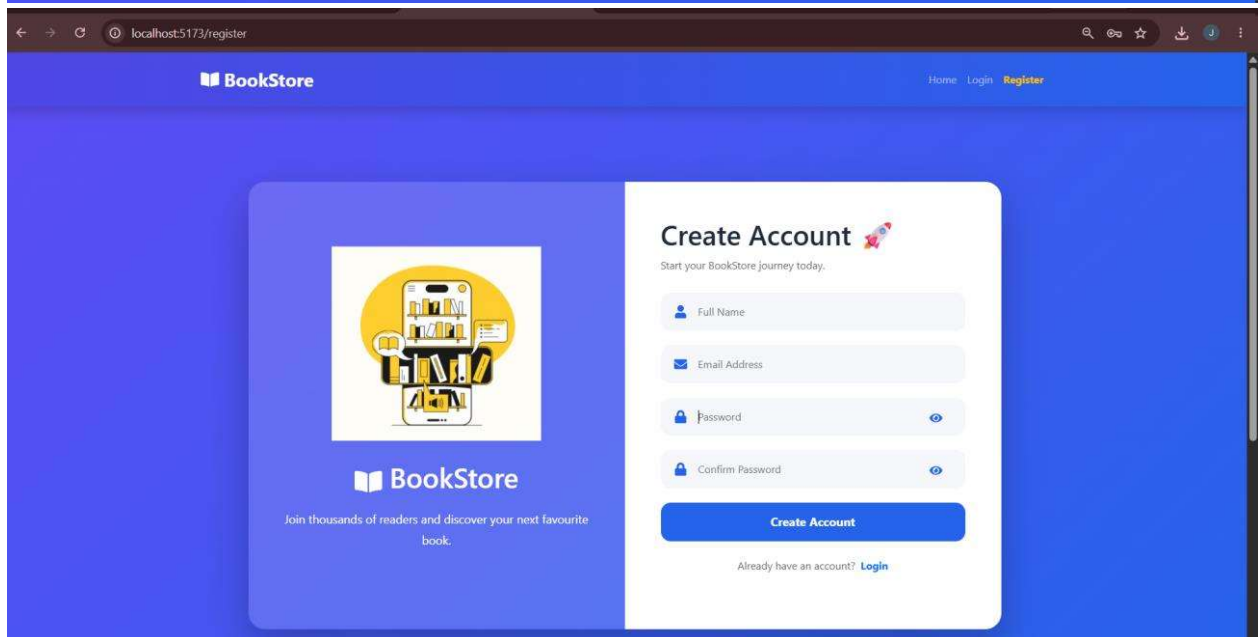
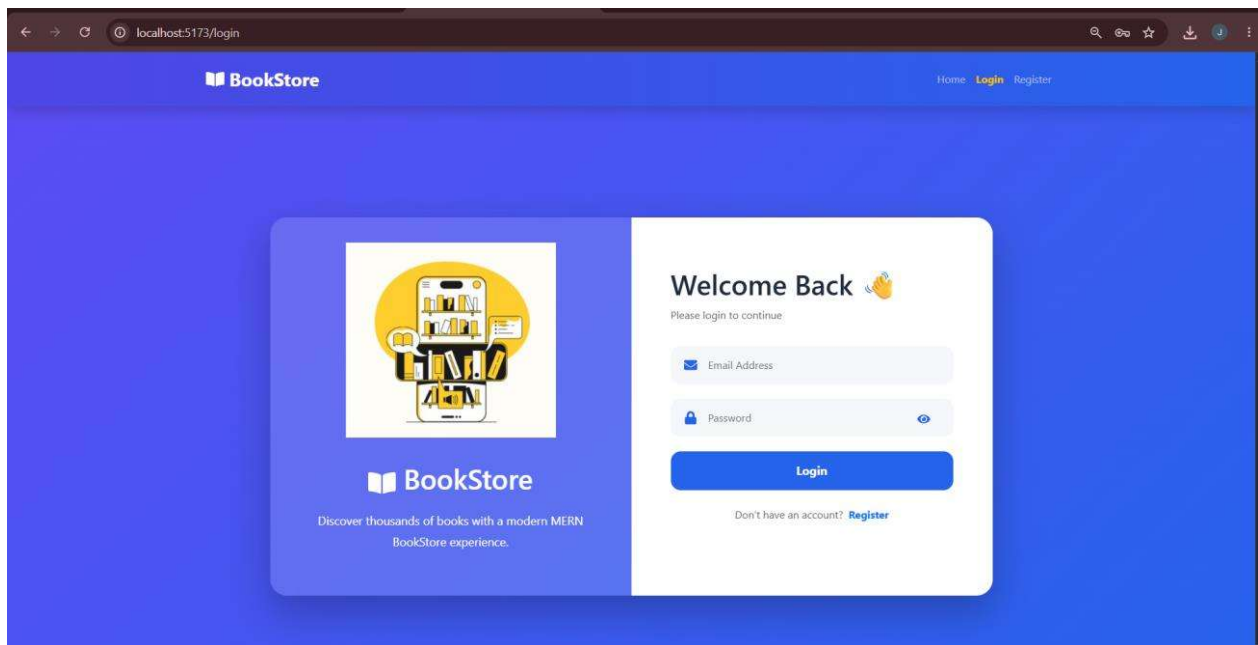
3

Categories



₹532

Average Price



cloud.mongodb.com/v2/6a21433a62e617d7de6d59b6#/explorer/6a45efdad165fb3838b44117/bookstore

ORGANIZATION
LAKIREDDY BALI RED...

PROJECT
Project 0

5

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookstore

books

carts

orders

users

local

sample_mflix

bookstore

Cluster0 > bookstore

View monitoring

Visualize your data

Create collection

Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
books	-	36.86 kB	703.00 B	3	234.00 B	1	36.86 kB
carts	-	36.86 kB	119.00 B	1	119.00 B	1	36.86 kB
orders	-	36.86 kB	504.00 B	2	252.00 B	1	36.86 kB
users	-	36.86 kB	416.00 B	2	208.00 B	2	73.73 kB

Alert System Status: All Good

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cloud.mongodb.com/v2/6a21433a62e617d7de6d59b6#/explorer/6a45efdad165fb3838b44117/bookstore/books/find

ORGANIZATION LAKIREDDY BALI RED... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookstore
 - books**
 - carts
 - orders
 - users
 - local
 - sample_mflix

Cluster0 > bookstore > books

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[View monitoring](#) [Visualize Your Data](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1-3 of 3

```
{ "_id": ObjectId('6a4baa394f53e8832fd067ce'),
  "title": "The Alchemist",
  "author": "Paulo Coelho",
  "price": 499,
  "category": "Novel",
  "description": "A famous inspirational novel.",
  "image": "/images/al-chemist.jpg",
  "createdAt": 2026-07-06T13:14:33.243+00:00,
  "updatedAt": 2026-07-06T13:15:14.324+00:00,
  "__v": 0 }
```

```
{ "_id": ObjectId('6a4bc19af1c781dd7c056c9f'),
  "title": "Atomic Habits",
  "author": "James Clear",
  "price": 599,
  "category": "Self-Help",
  "createdAt": 2026-07-06T13:14:33.243+00:00,
  "updatedAt": 2026-07-06T13:15:14.324+00:00,
  "__v": 0 }
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a21433a62e617d7de6d59b6#/explorer/6a45efdad165fb3838b44117/bookstore/carts/find

ORGANIZATION LAKIREDDY BALI RED... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookstore
 - books
 - carts**
 - orders
 - users
 - local
 - sample_mflix

Cluster0 > bookstore > carts

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[View monitoring](#) [Visualize Your Data](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1-1 of 1

```
{ "_id": ObjectId('6a4bc221f1c781dd7c056cb1'),
  "user": ObjectId('6a4ba2a77ed9df5710313a54'),
  "book": ObjectId('6a4baa394f53e8832fd067ce'),
  "quantity": 1,
  "createdAt": 2026-07-06T14:56:33.141+00:00,
  "updatedAt": 2026-07-06T14:56:33.141+00:00,
  "__v": 0 }
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a21433a62e617d7de6d59b6#/explorer/6a45efdad165fb3838b44117/bookstore/orders/find

ORGANIZATION LAKIREDDY BALI RED... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookstore
 - books
 - carts
 - orders
 - users
- local
- sample_mflix

orders

Cluster0 > bookstore > orders

Documents 2 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1-2 of 2

```
{
  "_id": ObjectId("6a4bb81b19ca8f42d40f2d9e"),
  "user": ObjectId("6a4ba2a77ed9df5710313a54"),
  "items": Array (1)
    totalPrice: 499
  "address": "Vijayawada"
  "paymentMethod": "Cash On Delivery"
  "status": "Pending"
  "createdAt": 2026-07-06T13:39:39.541+00:00
  "updatedAt": 2026-07-06T13:39:39.541+00:00
  "...v": 0
}
```

```
{
  "_id": ObjectId("6a4bc4ebf1c781dd7c056ce7"),
  "user": ObjectId("6a4bc42bf1c781dd7c056cbc"),
  "items": Array (1)
    totalPrice: 599
  "address": "Vijayawada"
}
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a21433a62e617d7de6d59b6#/explorer/6a45efdad165fb3838b44117/bookstore/orders/find

ORGANIZATION LAKIREDDY BALI RED... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookstore
 - books
 - carts
 - orders
 - users
- local
- sample_mflix

orders

Cluster0 > bookstore > orders

Documents 2 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1-2 of 2

```
{
  "status": "Pending"
  "createdAt": 2026-07-06T13:39:39.541+00:00
  "updatedAt": 2026-07-06T13:39:39.541+00:00
  "...v": 0
}
```

```
{
  "_id": ObjectId("6a4bc4ebf1c781dd7c056ce7"),
  "user": ObjectId("6a4bc42bf1c781dd7c056cbc"),
  "items": Array (1)
    totalPrice: 599
  "address": "Vijayawada"
  "paymentMethod": "Cash On Delivery"
  "status": "Pending"
  "createdAt": 2026-07-06T15:08:27.147+00:00
  "updatedAt": 2026-07-06T15:08:27.147+00:00
  "...v": 0
}
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a21433a62e617d7de6d59b6#/explorer/6a45efdad165fb3838b44117/bookstore/users/find

ORGANIZATION LAKIREDDY BALI RED... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookstore
 - books
 - carts
 - orders
 - users
 - local
 - sample_mflix

Cluster0 > bookstore > users

Documents 2 Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1-2 of 2

```
{
  "_id": ObjectId("6a4ba2a77ed0d5710313a54"),
  "name": "Shaik Jaherun",
  "email": "shaik@gmail.com",
  "password": "$2b$10$IhNTXQpke09R0REX3XuR4esGxU0b5mhYDjXdYeCjs2w7u09g3w6FC",
  "role": "admin",
  "createdAt": 2026-07-06T12:42:15.300+00:00,
  "updatedAt": 2026-07-06T12:42:15.300+00:00,
  "__v": 0
}
```

```
{
  "_id": ObjectId("6a4bc42bf1c781dd7c056cbc"),
  "name": "Jama1",
  "email": "jama112@gmail.com",
  "password": "$2b$10$Zy7lSmeVewt2Hfp3CePs5.MH84W0xY5luq1dfttB16J36hhv6j25u",
  "role": "admin",
  "createdAt": 2026-07-06T15:05:15.506+00:00,
  "updatedAt": 2026-07-06T15:05:15.506+00:00,
  "__v": 0
}
```

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Documents 2 Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1-2 of 2

```
{
  "email": "shaik@gmail.com",
  "password": "$2b$10$IhNTXQpke09R0REX3XuR4esGxU0b5mhYDjXdYeCjs2w7u09g3w6FC",
  "role": "admin",
  "createdAt": 2026-07-06T12:42:15.300+00:00,
  "updatedAt": 2026-07-06T12:42:15.300+00:00,
  "__v": 0
}
```

```
{
  "_id": ObjectId("6a4bc42bf1c781dd7c056cbc"),
  "name": "Jama1",
  "email": "jama112@gmail.com",
  "password": "$2b$10$Zy7lSmeVewt2Hfp3CePs5.MH84W0xY5luq1dfttB16J36hhv6j25u",
  "role": "admin",
  "createdAt": 2026-07-06T15:05:15.506+00:00,
  "updatedAt": 2026-07-06T15:05:15.506+00:00,
  "__v": 0
}
```

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DEMO LINK:

<https://drive.google.com/file/d/1p9Y6rpYOKavatTgB66eMDlOvTMAiJjao/view?usp=sharing>

12. KNOWN ISSUES

Currently, the application does not include online payment integration or email notifications. These features can be added in future versions. Apart from these enhancements, all implemented functionalities work as expected.

13. FUTURE ENHANCEMENTS

- Online Payment Gateway (Razorpay/Stripe)
- Email Notifications
- Wishlist Feature
- Book Reviews & Ratings
- Admin Analytics Dashboard
- AI-Based Book Recommendation
- Multi-language Support
- Dark Mode
- Inventory Management
- Invoice Generation